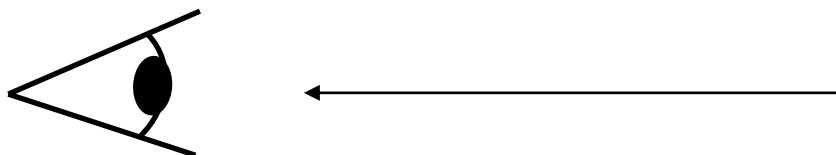


What is light?

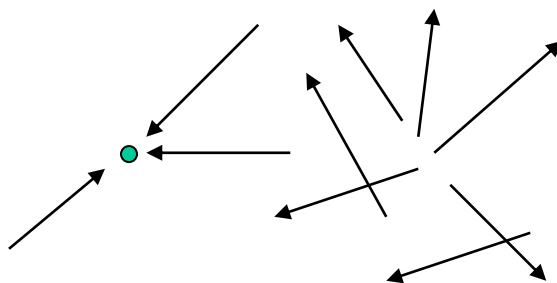
Electromagnetic radiation (EMR) moving along rays in space

- $R(\lambda)$ is EMR, measured in units of power (watts)
 - λ is wavelength



Light field

- We can describe all of the light in the scene by specifying the radiation (or “**radiance**” along all light rays) arriving at every point in space and from every direction



$$R(X, Y, Z, \theta, \phi, \lambda, t)$$

The light field

$$R(X, Y, Z, \theta, \phi, \lambda, t)$$

- Known as the **plenoptic function**
- If you know R , you can predict how the scene would appear from any viewpoint.

Additional Reading

The plenoptic function

- **Original reference:** E. Adelson and J. Bergen, "[The Plenoptic Function and the Elements of Early Vision](#)," in M. Landy and J. A. Movshon, (eds) Computational Models of Visual Processing, MIT Press 1991.
- L. McMillan and G. Bishop, "[Plenoptic Modeling: An Image-Based Rendering System](#)", Proc. SIGGRAPH, 1995, pp. 39-46.

The light field

- M. Levoy and P. Hanrahan, "[Light Field Rendering](#)", Proc SIGGRAPH 96, pp. 31-42.
- S. J. Gortler, R. Grzeszczuk, R. Szeliski, and M. F. Cohen, "[The lumigraph](#)," in Proc. SIGGRAPH, 1996, pp. 43-54.

The interaction of light and matter

What happens when a light ray hits a point on an object?

- Some of the light gets absorbed
 - converted to other forms of energy (e.g., heat)
- Some gets transmitted through the object
 - possibly bent, through “refraction”
- Some gets reflected
 - as we saw before, it could be reflected in multiple directions at once

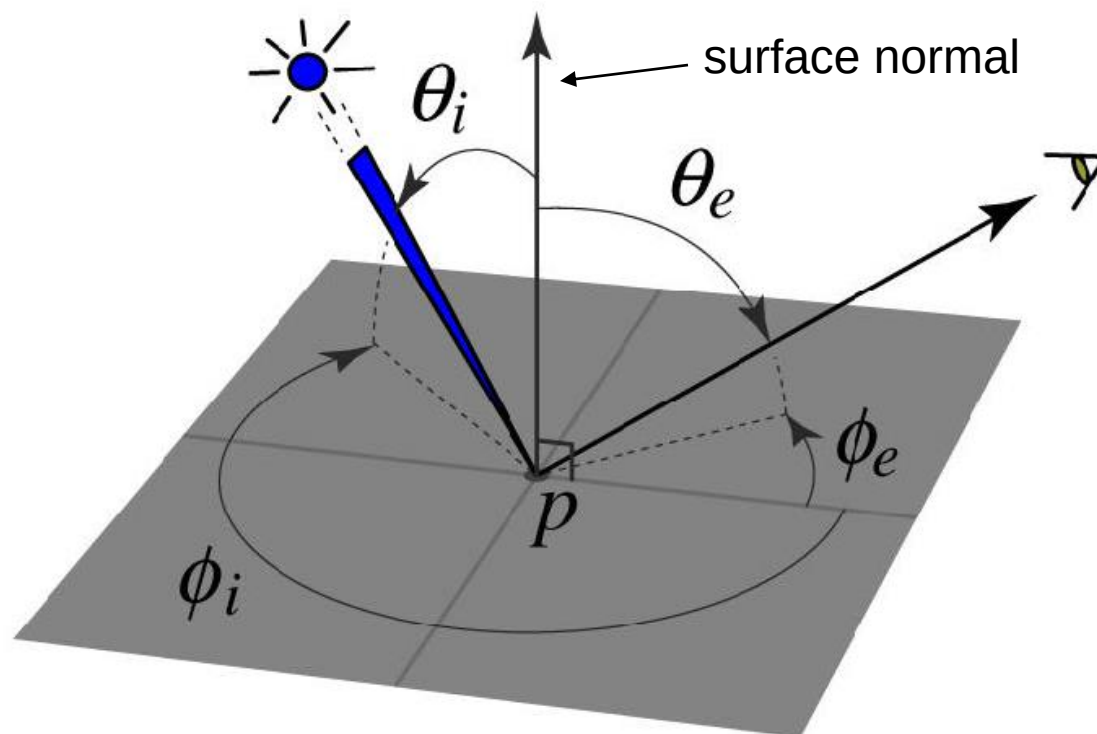
Let's consider the case of reflection in detail

- In the most general case, a single incoming ray could be reflected in all directions. How can we describe the amount of light reflected in each direction?

The BRDF

The Bidirectional Reflection Distribution Function

- Given an incoming ray (θ_i, ϕ_i) and outgoing ray (θ_e, ϕ_e)
what proportion of the incoming light is reflected along outgoing ray?



Answer given by the BRDF: $\rho(\theta_i, \phi_i, \theta_e, \phi_e)$

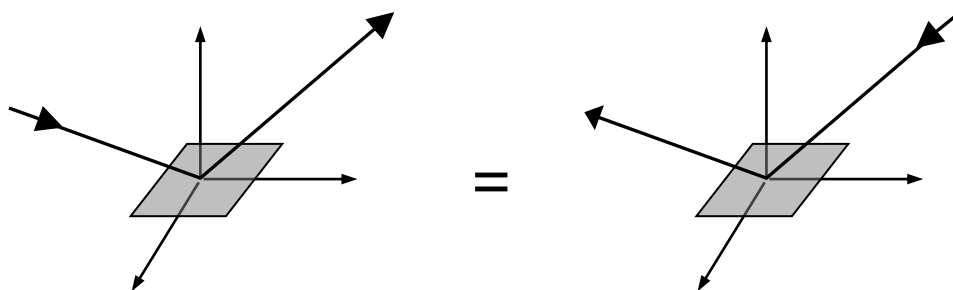
Constraints on the BRDF

Energy conservation

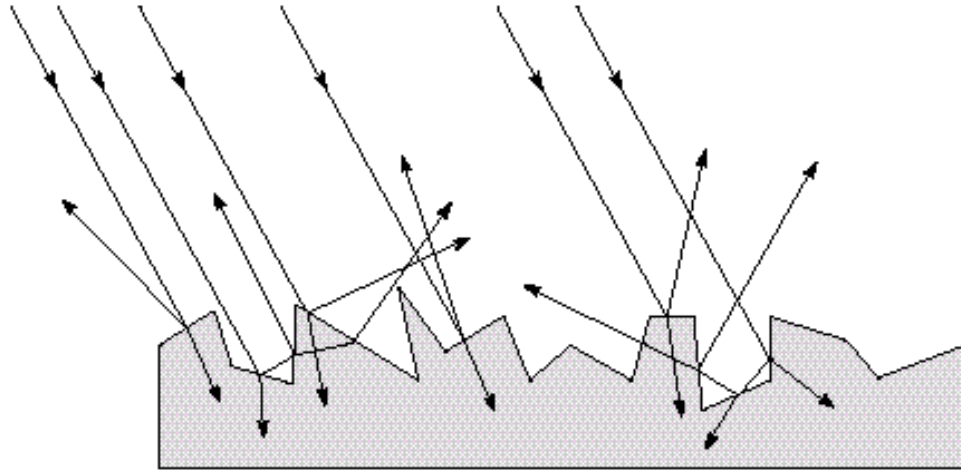
- Quantity of outgoing light $\leq$ quantity of incident light
 - integral of BRDF ≤ 1

Helmholtz reciprocity

- reversing the path of light produces the same reflectance



Diffuse reflection



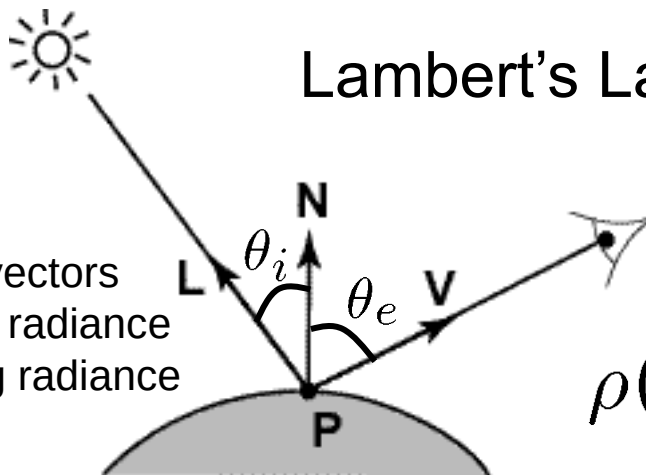
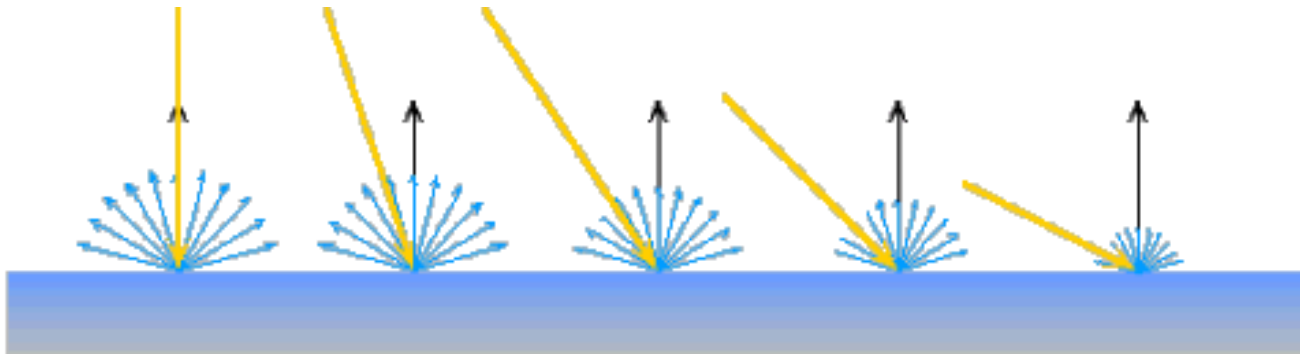
Diffuse reflection

- Dull, matte surfaces like chalk or latex paint
- Microfacets scatter incoming light randomly
- Effect is that light is reflected equally in all directions

Diffuse reflection

Diffuse reflection governed by **Lambert's law**

- Viewed brightness does not depend on viewing direction
- Brightness *does* depend on direction of illumination
- This is the model most often used in computer vision



$\mathbf{L}$, $\mathbf{N}$, $\mathbf{V}$ unit vectors
 I_e = outgoing radiance
 I_i = incoming radiance

Lambert's Law: $I_e = k_d \mathbf{N} \cdot \mathbf{L} I_i$

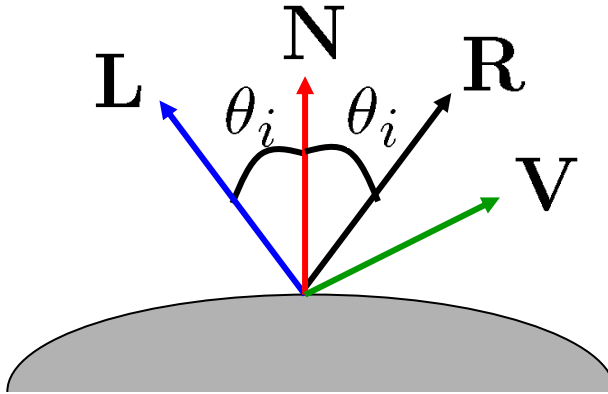
k_d is called **albedo**

BRDF for **Lambertian surface**

$$\rho(\theta_i, \phi_i, \theta_e, \phi_e) = k_d \cos \theta_i$$

Specular reflection

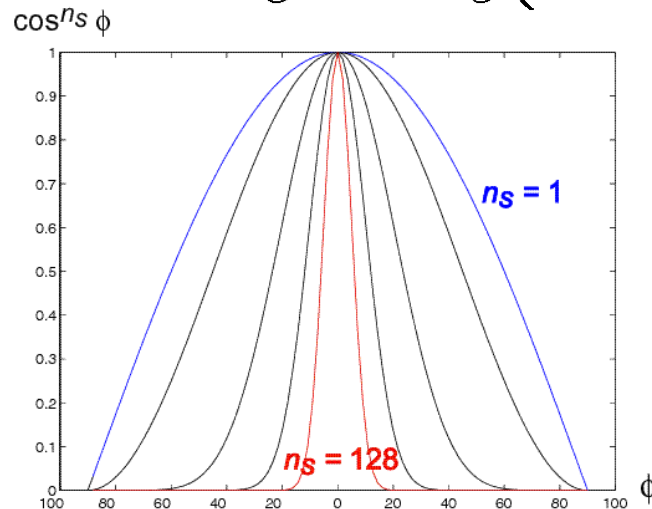
For a perfect mirror, light is reflected about **N**



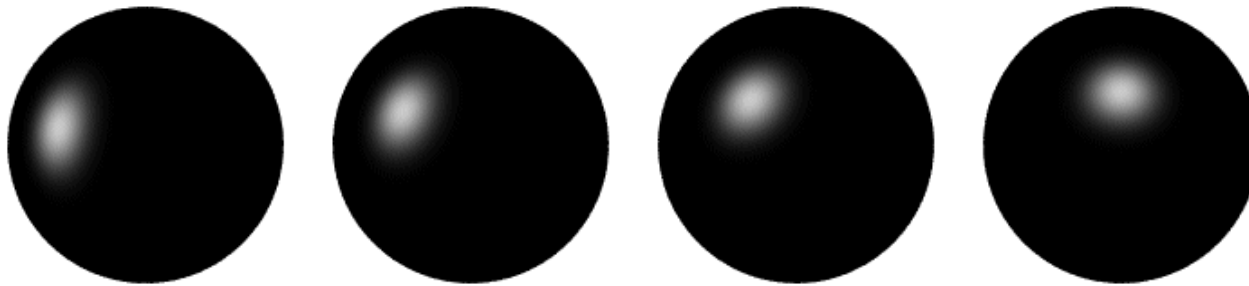
$$I_e = \begin{cases} I_i & \text{if } \mathbf{V} = \mathbf{R} \\ 0 & \text{otherwise} \end{cases}$$

Near-perfect mirrors have a **highlight** around **R**

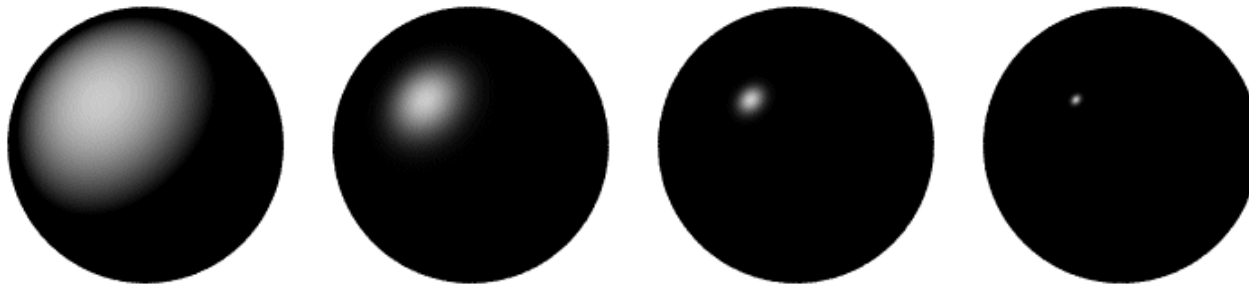
- common model: $I_e = k_s (\mathbf{V} \cdot \mathbf{R})^{n_s} I_i$



Specular reflection



Moving the light source



Changing n_s

Phong illumination model

Phong approximation of surface reflectance

- Assume reflectance is modeled by three components
 - Diffuse term
 - Specular term
 - Ambient term (to compensate for inter-reflected light)

$$I_e = k_a I_a + I_i \left[k_d (\mathbf{N} \cdot \mathbf{L})_+ + k_s (\mathbf{V} \cdot \mathbf{R})_+^{n_s} \right]$$

$\mathbf{L}$, $\mathbf{N}$, $\mathbf{V}$ unit vectors

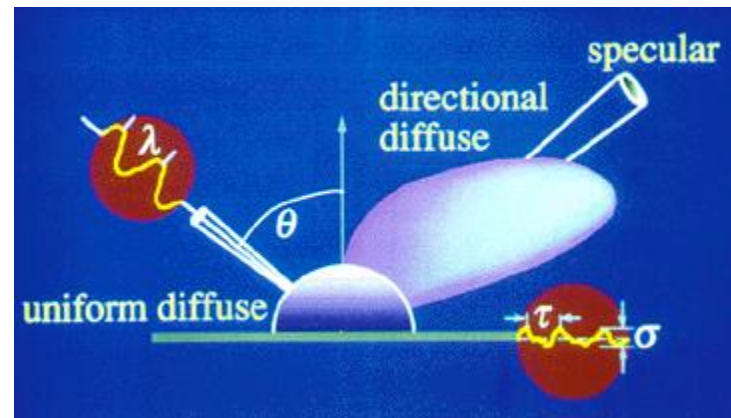
I_e = outgoing radiance

I_i = incoming radiance

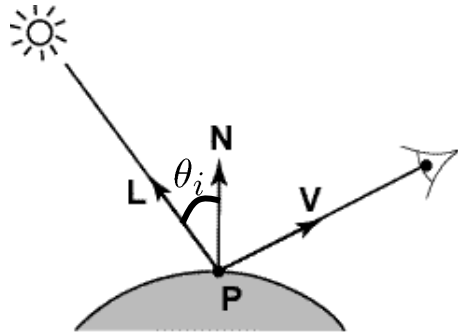
I_a = ambient light

k_a = ambient light reflectance factor

$(x)_+ = \max(x, 0)$



Shape from shading



Suppose $k_d = 1$

$$\begin{aligned} I &= k_d \mathbf{N} \cdot \mathbf{L} \\ &= \mathbf{N} \cdot \mathbf{L} \\ &= \cos \theta_i \end{aligned}$$

You can directly measure angle between normal and light source

- Not quite enough information to compute surface shape
- But can be if you add some additional info, for example
 - assume a few of the normals are known (e.g., along silhouette)
 - constraints on neighboring normals—“integrability”
 - smoothness
- Hard to get it to work well in practice
 - plus, how many real objects have constant albedo?